(see pp.20–21). For the first time, people had the skills to live in harsh environments like the Eurasian steppes, where there is little rainfall and dramatic changes in temperature with hot summers and very cold winters. Despite these skills, the Cro-Magnons appear to have moved south into sheltered locations, only moving north again as temperatures rose. Some of them constructed elaborate dwellings, like the intricate mammoth bone houses at Mezhirich in modern Ukraine (see left), built partially into the ground and roofed with hides and sod. Toward the end of the Ice Age, about 12,000 years ago, human society became more elaborate, as populations grew larger and new areas were colonized.

Siberia and the tundra

Homo sapiens migrated north from southwestern Asia and colonized the river valleys of Central Asia around 45,000 years ago. Small bands lived permanently in the bitter cold of the steppe-tundra—a windswept landscape featuring low-growing vegetation—that extended from central Europe all the way to Siberia far to the northeast. Enduring long winters, each band anchored itself on shallow river valleys like those of the Don and Dnieper in Russia, subsisting for the most part on animals such as the saiga antelope and large game, including the arctic elephant and the mammoth.

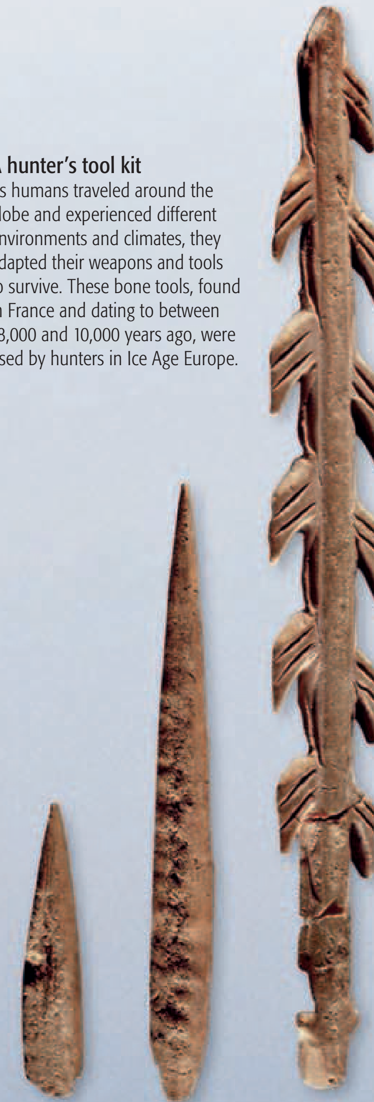
HOW WE KNOW

ADAPTING TO CHANGE

Study of the genes of modern populations can help to show how the early humans colonized the planet. Mitochondrial DNA, inherited through the maternal line back to a fictional "Eve" (see p.24), can be traced from an ancestral tropical African population to today. The male Y chromosome can also be used to trace through generations. From this evidence we know that 99.9 percent of the genetic code of modern humans is identical throughout the world. The differences in facial features and coloring are down to minor genetic mutations that have taken place over the last 150,000 years. Amazingly, the world's population outside Africa can trace their genetic history back to perhaps as few as 1,000 individuals who made the journey out of that continent. Chromosome mutations can be used to show when groups arrived in different parts of the world and to construct a genetic family tree that goes back to the Ice Age.

Between 35,000 and 18,000 years ago, some hunting bands moved northeastward across the steppe-tundra into the Lake Baikal region of Siberia and farther to the northeast. Some moved to, or formed, new groups, while others moved to find new hunting grounds or natural resources. A variety of circumstances linked to hunting and survival contributed to the movement of tiny numbers of these late Ice Age bands across an extremely inhospitable landscape. Such natural population movements led to vast areas of the globe being colonized.

Even earlier, from around *c.* 60,000 years ago, other groups moved east from northeast Africa and southwestern Asia into what is now India and Pakistan, and into the tropical forests of Southeast Asia. We know little of these movements—the groups probably skirted the Eurasian »



A hunter's tool kit
As humans traveled around the globe and experienced different environments and climates, they adapted their weapons and tools to survive. These bone tools, found in France and dating to between 18,000 and 10,000 years ago, were used by hunters in Ice Age Europe.